

145. 2. $A(2,1,0)$ $B(1,3,-2)$

5. Puntak
Zunyak
etc. plawak

Avikitu $\overline{AP} = \frac{1}{3} \overline{AB}$ eta $\overline{AQ} = \frac{2}{3} \overline{AB}$ beteko
daben P eta Q puntak.

$$\overline{AB} = \overline{OB} - \overline{OA} = (1, 3, -2) - (2, 1, 0) = (-1, 2, -2)$$

$$\overline{AP} = \frac{1}{3} \overline{AB} = \frac{1}{3} (-1, 2, -2)$$

$$\overline{AQ} = \frac{2}{3} \overline{AB} = \frac{2}{3} (-1, 2, -2)$$

$$\begin{aligned} \overline{AQ} &= \overline{OQ} - \overline{OA} \rightarrow \overline{OQ} = \overline{AQ} + \overline{OA} = \\ &= \frac{2}{3} (-1, 2, -2) + (2, 1, 0) \\ &= \left(-\frac{2}{3} + 2, \frac{4}{3} + 1, -\frac{4}{3} + 0 \right) \end{aligned}$$

$$\boxed{\overline{OQ} = \left(\frac{4}{3}, \frac{7}{3}, -\frac{4}{3} \right)}$$

$$\overline{AP} = \overline{OP} - \overline{OA} = \frac{1}{3} (-1, 2, -2)$$

$$\overline{OP} = \overline{AP} + \overline{OA} = \frac{1}{3} (-1, 2, -2) + (2, 1, 0)$$

$$\boxed{\overline{OP} = \left(\frac{5}{3}, \frac{5}{3}, -\frac{2}{3} \right)}$$

145. 3/ a-reu eta b-reu balizak $A(1,2,-1)$, $B(3,0,-2)$
eta $C(4,a,b)$ lorokatu egoteko.

Lorokatu egoteko $\overline{AB}$ eta $\overline{BC}$ proportioan
izan behar dira

$$\overline{AB} = \overline{OB} - \overline{OA} = (3, 0, -2) - (1, 2, -1) = (2, -2, -1)$$

$$\overline{BC} = \overline{OC} - \overline{OB} = (4, a, b) - (3, 0, -2) = (1, a, b+2)$$

$$\frac{2}{1} = \frac{-2}{a} = \frac{-1}{b+2}$$

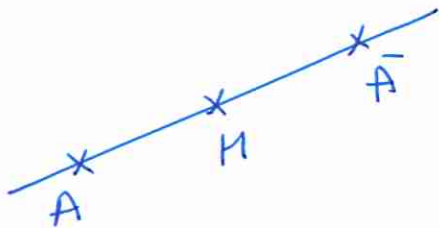
$$\rightarrow \boxed{\begin{matrix} a = -1 \\ b = -5/2 \end{matrix}}$$

$$\begin{aligned} -2(b+2) &= -a \\ -2b-4 &= 1 \end{aligned}$$

145, 5 | A-reu simetrikoa π -rekiko

$$A(-2, 3, 0)$$

$$H(1, -1, 2)$$



$$\overline{AH} = \overline{HA'}$$

$$\overline{AA'} = 2\overline{AH}$$

$$\overline{AH} = \overline{OH} - \overline{OA} = (1, -1, 2) - (-2, 3, 0) \\ = (3, -4, 2).$$

$$2\overline{AH} = 2(3, -4, 2) = (6, -8, 4)$$

$$\overline{AA'} = 2\overline{AH} = (6, -8, 4).$$

$$\overline{AA'} = \overline{OA'} - \overline{OA} = (6, -8, 4).$$

$$\overline{OA'} = (6, -8, 4) + \overline{OA} = (6, -8, 4) + (-2, 3, 0)$$

$$\boxed{\overline{OA'} = (4, -5, 4)}$$

Beste modu batena :

$$H = \left(\frac{x_A + x_{A'}}{2}, \frac{y_A + y_{A'}}{2}, \frac{z_A + z_{A'}}{2} \right)$$

$$(1, -1, 2) = \left(\frac{-2 + x_{A'}}{2}, \frac{3 + y_{A'}}{2}, \frac{0 + z_{A'}}{2} \right)$$

$$\left\{ \begin{array}{l} 1 = \frac{-2 + x_{A'}}{2} \longrightarrow x_{A'} = 4. \\ -1 = \frac{3 + y_{A'}}{2} \longrightarrow y_{A'} = -5 \\ 2 = \frac{0 + z_{A'}}{2} \longrightarrow z_{A'} = 4. \end{array} \right\} \boxed{H(4, -5, 4)}$$

ZUZENAK

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koordinatu ardazko parametrikak eta implizituak

OX ardazko

$$\vec{a}_r(1,0,0)$$

$$P(0,0,0)$$

PARAMET.

$$\begin{cases} x=1 \\ y=0 \\ z=0 \end{cases}$$

INPLIZ.

$$\begin{cases} y=0 \\ z=0 \end{cases}$$

OY ardazko

$$\vec{a}_r(0,1,0)$$

$$P(0,0,0)$$

$$\begin{cases} x=0 \\ y=1 \\ z=0 \end{cases}$$

$$\begin{cases} x=0 \\ z=0 \end{cases}$$

OZ ardazko

$$\vec{a}_r(0,0,1)$$

$$P(0,0,0)$$

$$\begin{cases} x=0 \\ y=0 \\ z=1 \end{cases}$$

$$\begin{cases} x=0 \\ y=0 \end{cases}$$

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$$A(-4, 2, 5)$$

OZ ardazkoarekin II. $\vec{a}_r(0,0,1)$

$$(x,y,z) = (-4, 2, 5) + \lambda(0,0,1)$$

BEKTORIALA

$$\begin{cases} x=-4 \\ y=2 \\ z=5+\lambda \end{cases} \quad \text{PARAMETRIKAK.}$$

$$x = -4 + \lambda \cdot 0 \rightarrow \lambda = \frac{x+4}{0}$$

$$y = 2 + \lambda \cdot 0 \rightarrow \lambda = \frac{y-2}{0}$$

$$z = 5 + \lambda \rightarrow \lambda = \frac{z-5}{1}$$

EK JARRAITZA

$$\frac{x+4}{0} = \frac{y-2}{0} = \frac{z-5}{1}$$

Implizituak.

$$\begin{cases} x+4=0 \\ y-2=0 \end{cases}$$

12 $\left\{ \begin{array}{l} P(1, -3, 0) \\ \vec{u} \wedge \vec{v} \text{ vektorekiko paralelo.} \end{array} \right.$

$$\vec{u}(1, -1, 2) \quad \vec{v}(2, 0, 0)$$

$$\vec{u} \wedge \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 0 & 0 \end{vmatrix} = 4\hat{j} + 2\hat{k} \rightarrow (0, 4, 2)$$

$$r: \left\{ \begin{array}{l} P(1, -3, 0) \\ \vec{dr}(0, 4, 2) \end{array} \right.$$

$$(x, y, z) = (1, -3, 0) + \lambda(0, 4, 2) \quad \text{BKT.}$$

PARA λ $\rightarrow \left\{ \begin{array}{l} x = 1 \\ y = -3 + 4\lambda \\ z = 2\lambda \end{array} \right.$

$$\frac{x-1}{0} = \frac{y+3}{4} = \frac{z-0}{2} \quad \text{JARRAIT}$$

$$\begin{aligned} (x-1) \cdot 4 &= 0 \\ (x-1) \cdot 2 &= 0 \\ (y+3) \cdot 2 &= 4(z-0) \\ \Rightarrow y+3 &= 2z \end{aligned} \rightarrow \left\{ \begin{array}{l} x-1=0 \\ y-2z=-3 \end{array} \right. \quad \text{INPLUZTA}$$

13 $r: \left\{ \begin{array}{l} 2x - y + z - 3 = 0 \\ x - y + z - 2 = 0 \end{array} \right.$

kalkulatu parametroak

Sistema ebazten dogu:

$$\left\{ \begin{array}{l} 2x - y + z = 3 \\ x - y + z = 2 \end{array} \right.$$

$$\left\{ \begin{array}{l} 2x - y = 3 - \lambda \\ x - y = 2 - \lambda \\ z = \lambda \end{array} \right.$$

$$r: \left\{ \begin{array}{l} x = 1 \\ y = \lambda - 1 \\ z = \lambda \end{array} \right.$$

$$s: \left\{ \begin{array}{l} x + 2y - z - 3 = 0 \\ 3y + 2z - 1 = 0 \end{array} \right.$$

$$\left(\begin{array}{cc|c} 2 & -1 & 3-\lambda \\ 1 & -1 & 2-\lambda \end{array} \right).$$

$$x = \frac{\begin{vmatrix} 3-\lambda & -1 \\ 2-\lambda & -1 \end{vmatrix}}{-1} = \frac{\cancel{3} - 3 + 2 - \cancel{2}}{-1} = 1$$

$$y = \frac{\begin{vmatrix} 2 & 3-\lambda \\ 1 & 2-\lambda \end{vmatrix}}{-1} = \frac{4 - 2\lambda - 3 + \lambda}{-1} = \frac{1 - \lambda}{-1} = \lambda - 1$$

$$z = \lambda$$

$$S: \begin{cases} x + 2y - z = 3 \\ 3y + 2z = 1 \end{cases}$$

$$x = \frac{\begin{vmatrix} 3-1 & 2 \\ 1-2 & 3 \end{vmatrix}}{3} = \frac{9-3-2+4}{3} = \frac{3}{3} = 1$$

$$\begin{cases} x + 2y = 3 - z \\ 3y = 1 - 2z \\ z = 1 \end{cases}$$

$$y = \frac{\begin{vmatrix} 1 & 3-1 \\ 0 & 1-2 \end{vmatrix}}{3} = \frac{1-2}{3} = -\frac{1}{3}$$

$$\begin{vmatrix} 1 & 2 & 1 & 3-1 \\ 0 & 3 & 1 & 1-2 \end{vmatrix}$$

$$r: \begin{cases} x = \frac{4+1}{3} \\ y = \frac{1-2}{3} \\ z = 1 \end{cases}$$

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$$a) \begin{cases} x - y = 0 \\ y + z = 2 \end{cases} \rightarrow \begin{aligned} &\vec{n}_1(1, -1, 0) \\ &\vec{n}_2(0, 1, 1) \end{aligned}$$

$$\vec{d} = \vec{n}_1 \wedge \vec{n}_2 = (1, -1, 0) \wedge (0, 1, 1) =$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = -\hat{i} + \hat{k} - \hat{j}$$

$$\vec{d} = (-1, -1, 1)$$

b) Parametrisation

$$r: \begin{cases} P \\ \vec{d} \end{cases}$$

2 unknown point but 3 eqs.

$$\text{Addition } x=0 \rightarrow y=0 \rightarrow z=2 \quad \underline{\underline{P(0,0,2)}}$$

$$\begin{cases} x = 0 - t \\ y = 0 - t \\ z = 2 + t \end{cases}$$

Planes
be kline
 $\vec{n}(a,b,c)$

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$$r: \frac{x}{2} = \frac{y+1}{-1} = z$$

$$\frac{x}{2} = z \rightarrow x = 2z$$

$$\frac{x}{2} = \frac{y+1}{-1} \rightarrow 2y+2+x=0$$

$$\begin{cases} x - 2z = 0 \\ x + 2y + 2 = 0 \end{cases}$$

ZU ZWEIEN POSITION ERLÄUTEREN

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$$r: \frac{x-1}{3} = \frac{y+2}{2} = \frac{z-1}{4}$$

$$P_r(1, -2, 1)$$

$$\vec{dr}(3, 2, 4)$$

a)

$$s: \frac{x+2}{-1} = \frac{y-3}{2} = \frac{z-2}{3}$$

$$P_s(-2, 3, 2)$$

$$\vec{ds}(-1, 2, 3)$$

• Norabide 11? $\frac{3}{-1} \neq \frac{2}{2} \neq \frac{4}{3} \rightarrow$ NORABIDE DES BERDINA

• Gurutzatzen edo eba ki?

$$\vec{P_s P_r} = (-2, 3, 2) - (1, -2, 1) = (-3, 5, 1)$$

~~bera~~

$$\begin{vmatrix} -3 & 5 & 1 \\ 3 & 2 & 4 \\ -1 & 2 & 3 \end{vmatrix} = -51 \neq 0$$

$$\vec{P_s P_r} = (-2, 3, 2) - (1, -2, 1) = (-3, 5, 1)$$

$$\begin{vmatrix} \vec{P_s P_r} \\ \vec{dr} \\ \vec{ds} \end{vmatrix} = \begin{vmatrix} -3 & 5 & 1 \\ 3 & 2 & 4 \\ -1 & 2 & 3 \end{vmatrix} = -51 \neq 0$$

Hiru bektoreak eta diru planukoak dira
bera plano desberdina da
GURUTZATZEN diru

(3)

$$b) \quad r: \frac{x-1}{-1} = \frac{y-1}{2} = \frac{z-2}{1} \quad s: \frac{x-4}{4} = \frac{y-4}{1} = \frac{z-5}{2}$$

$$P_r(1, 1, 2) \\ \vec{d}_r(-1, 2, 1)$$

$$P_s(4, 4, 5) \\ \vec{d}_s(4, 1, 2)$$

• $\vec{d}_r$ eto $\vec{d}_s$ reu norabideak:

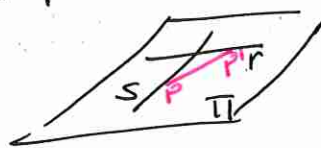
$$-\frac{1}{4} \neq \frac{2}{1} \neq \frac{1}{2} \Rightarrow \text{Norabide desberdino} \begin{cases} \text{Guzutatu} \\ \text{edo} \\ \text{Ebaki} \end{cases}$$

• $\vec{P}_r \vec{P}_s$ bektorea:

$$\vec{P}_r \vec{P}_s = (4, 4, 5) - (1, 1, 2) = (3, 3, 3)$$

Plaukideok diru $\vec{d}_r, \vec{d}_s$ eto $\vec{P}_r \vec{P}_s$?!

$$\begin{vmatrix} -1 & 2 & 1 \\ 4 & 1 & 2 \\ 3 & 3 & 3 \end{vmatrix} = 0 \rightarrow \text{Plaukideok diru}$$



ZUZENAK EBAKITUEN
DIRA

$$c) r: \frac{x}{2} = y-1 = \frac{z+1}{3}$$

$$P_r(0, 1, -1) \\ \vec{d}_r(2, 1, 3)$$

$$s: \begin{cases} x-2y-1=0 \\ 3y-z+1=0 \end{cases} \begin{matrix} x=0 \\ y=-1/2 \\ -3/2 - z + 1 = 0 \\ z = -1/2 \end{matrix}$$

$$P_s(0, -\frac{1}{2}, -\frac{1}{2})$$

$$\vec{d}_s = \vec{n}_1 \wedge \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 0 \\ 0 & 3 & -1 \end{vmatrix}$$

$$2\hat{i} + 3\hat{k} + \hat{j} \\ \vec{d}_s = (2, 1, 3)$$

• $\vec{d}_r$ eto $\vec{d}_s$ reu norabideak!

$$\frac{2}{2} = \frac{1}{1} = \frac{3}{3}$$

Bardinok
diraue?

koitidentetuek
edo
paralelek.

• $\vec{P}_r \in r$ daua aztertuta ea S-u dojoan

$$P_r(0, 1, -1) \rightarrow s: \begin{cases} 0-2 \cdot 1-1 \neq 0 \\ 3 \cdot 1 - (-1) + 1 \neq 0 \end{cases} \text{Et do betituen}$$

ZUZENAK PARALELOAK diru

$$d) \quad r: \frac{x-1}{2} = \frac{y}{3} = \frac{z}{4}$$

$$P_r(1, 0, 0)$$

$$\vec{d}_r(2, 3, 4)$$

$$s: \begin{cases} x = 3 + 4t \\ y = 3 + 6t \\ z = 4 + 8t \end{cases}$$

$$P_s(3, 3, 4)$$

$$\vec{d}_s(4, 6, 8)$$

Notabideko atartuta:

~~$$\frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$~~

$$\frac{2}{4} = \frac{3}{6} = \frac{4}{8}$$

$\vec{d}_r$ eta $\vec{d}_s$
proportionalak
dira $\rightarrow$

Notabide berdira

Zuzenak || edo koinkidentak

dira beraz atartuta dogu ea $P_s \in r$

$$P_s(3, 3, 4) \rightarrow r \left\{ \frac{3-1}{2} = \frac{3}{2} = \frac{4}{4} \checkmark \right.$$

$P_s \in r \rightarrow$ ZUZENA KOINKIDENTAK DIRA

$$\frac{2(x-1/2)}{3} = \frac{x-1/2}{3/2}$$

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$$r: x=y=z=a$$

$$s: \frac{2x-1}{3} = \frac{y+3}{-2} = \frac{z-2}{0}$$

Erakitzeko $\vec{d}_r \times \vec{d}_s$ eta $\vec{d}_r, \vec{d}_s$ eta $\vec{P_r P_s}$
planokideak izan behar dira, beraz euren
determinantea = 0.

$$\vec{d}_r = (1, 1, 1)$$

$$P_r = (0, 0, a)$$

$$\vec{d}_s = (3/2, -2, 0)$$

$$P_s = (1/2, -3, 2)$$

$$\vec{P_r P_s} = \left(\frac{1}{2}, -3, 2 \right) - (0, 0, a) = \left(\frac{1}{2}, -3, 2-a \right)$$

$$\begin{vmatrix} \frac{1}{2} & -3 & 2-a \\ 1 & 1 & 1 \\ 3/2 & -2 & 0 \end{vmatrix} = \frac{-9}{2} - 2(2-a) - \frac{3}{2}(2-a) + 1 =$$

$$= \frac{-9}{2} - 4 + 2a - 3 + \frac{3a}{2} + 1 = \frac{5a}{2} - \frac{21}{2} = 0$$

$$a = 3$$

Ebakite puntus kalkulatzeko $a=3$ dauka: ④

$$r: x=y=z-3$$

$$s: \frac{2x-1}{3} = \frac{y+3}{-2} = \frac{z-2}{0}$$

PARAMETRIZETARA pasatuz:

$$r: \begin{cases} x = \lambda \\ y = \lambda \\ z = \lambda + 3 \end{cases}$$

$$s: \begin{cases} \frac{2x-1}{3}: \mu & x = \frac{1}{2} + \frac{3}{2}\mu \\ \frac{y+3}{-2}: \mu & y = -3 - 2\mu \\ \frac{z-2}{0}: \mu & z = 2 \end{cases}$$

$$\begin{array}{lcl} x \text{ berdinduz} \rightarrow & \lambda = \frac{1}{2} + \frac{3}{2}\mu & \\ y \text{ " } \rightarrow & \lambda = -3 - 2\mu & \\ z \text{ " } \rightarrow & \lambda + 3 = 2 & \end{array} \left\{ \begin{array}{l} \lambda - \frac{3}{2}\mu = \frac{1}{2} \\ \lambda + 2\mu = -3 \\ \lambda = -1 \end{array} \right.$$

$$\lambda = -1 \left\{ \begin{array}{l} \mu = \left(\frac{-3+1}{2}\right) = -1 \\ \mu = \left(\frac{1}{2} + 1\right)\left(-\frac{2}{3}\right) = \frac{3}{2}\left(-\frac{2}{3}\right) = -1 \end{array} \right.$$

$$\boxed{\lambda = -1 \rightarrow \mu = -1} \rightarrow r \text{ eta } s - u.$$

$$r: \begin{cases} x = -1 \\ y = -1 \\ z = 2 \end{cases}$$

$$s: \begin{cases} x = \frac{1}{2} - \frac{3}{2} = -1 \\ y = -3 - 2(-1) = -1 \\ z = 2 \end{cases}$$

$$\boxed{P(-1, -1, 2)}$$

RI ZUZENEN BERTID ERLANBOA

1.) KOMPATATU ea $\vec{dr}$ eta $\vec{ds}$ NORABIDE BERIDEXKOK DIREN

$$\frac{dr_x}{ds_x} = \frac{dr_y}{ds_y} = \frac{dr_z}{ds_z}$$

NORABIDE BERDINA $\leq$

• Propotionalak badina.

• Aztertuta ea $P_r \in s$.

$P_r \in s$ —
 $P_r \notin s$ =

NORABIDE DIBERDINTA

Aztertuta ea ebaki
 ala gurutzatu diren.

(hirugarren bektoreak) alaba $\vec{P_r P_s}$

Aztertuta ea planokideak diren $\vec{dr}, \vec{ds}$ $\vec{P_r P_s}$



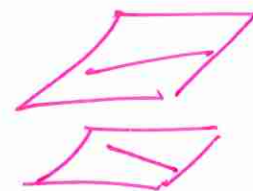
$$\begin{vmatrix} dr_x & dr_y & dr_z \\ ds_x & ds_y & ds_z \\ \vec{P_r P_s}_x & \vec{P_r P_s}_y & \vec{P_r P_s}_z \end{vmatrix}$$

$= 0$ Planokideak



$\neq 0$

Plano
 desberdinak
 GURUTZATU



18 Kalkulatu m paralelboak izateko:

(5)

$$r: \begin{cases} x = 5 + 4t \\ y = 3 + t \\ z = -t \end{cases}$$

$$s: \frac{x}{m} = \frac{y-1}{3} = \frac{z+3}{n}$$

$$P(5, 3, 0) \\ \vec{dr} (4, 1, -1)$$

$$P(0, 1, -3) \\ \vec{ds} (m, 3, n)$$

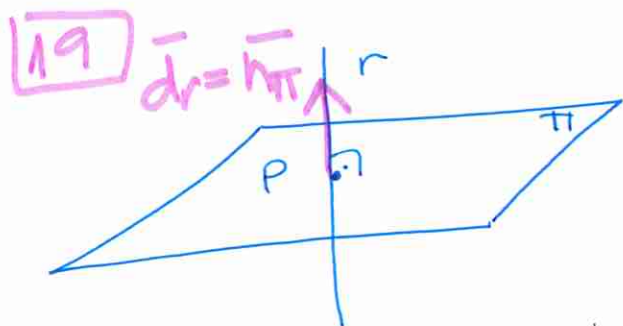
• PARALELOAK BADIRA $\rightarrow \vec{dr}$ eta $\vec{ds}$ posizioak diren.

$$\frac{4}{m} = \frac{1}{3} = \frac{-1}{n} \Rightarrow \boxed{\begin{matrix} m = 12 \\ n = -3 \end{matrix}}$$

• P_r alderatzen da ea $\in s$ dagoen.

$$P(5, 3, 0) \rightarrow \frac{5}{12} \neq \frac{3-1}{3} \neq \frac{0+3}{-3} \quad \text{Beraz } \boxed{P_r \notin s}$$

PARALELOAK dira



$$\pi: x + 2y = 5 \rightarrow \vec{n}_\pi = (1, 2, 0) \\ P(7, -2, 1)$$

Kalkulatzeko π planoarekiko zuzen perpendikularra bere puntu eta $\vec{dr}$ behar da (π planoareko normala).

$$P(7, -2, 1) \\ \vec{dr} = \vec{n}_\pi = (1, 2, 0)$$

$$r: \begin{cases} x = 7 + t \\ y = -2 + 2t \\ z = 1 \end{cases} \Rightarrow \frac{x-7}{1} = \frac{y+2}{2} = \frac{z-1}{0}$$

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$$S: \begin{cases} P(-1, 5, 0) \\ r \text{ zuzuenen paralelo} \end{cases} \quad r: \begin{cases} x+y=1 \rightarrow \vec{n}_1=(1,1,0) \\ y-z=2 \rightarrow \vec{n}_2=(0,1,-1) \end{cases}$$

S zuzuenaren norabide bektorea $\vec{ds}$, $\vec{dr}$ -ren
bardina da paraleloa direlako.

$$\begin{aligned} \vec{dr} &= \vec{n}_1 \wedge \vec{n}_2 = (1, 1, 0) \wedge (0, 1, -1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \\ &= -\hat{i} + \hat{k} + \hat{j} = (-1, 1, 1) \end{aligned}$$

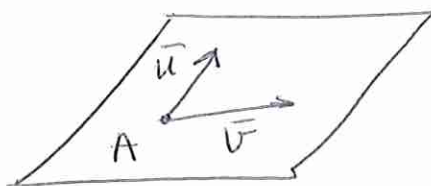
$$\text{Beraz } S \text{ zuzeno } \begin{cases} P(-1, 5, 0) \\ \vec{ds} = (-1, 1, 1) \end{cases}$$

$$S: \begin{cases} x = -1 + \lambda \\ y = 5 + \lambda \\ z = 0 + \lambda \end{cases} \Rightarrow \frac{x+1}{-1} = \frac{y-5}{1} = \frac{z}{1}$$

PLANOAK

21) INPLIZITAK

a/ $A(1, -3, 2)$
 $\vec{u}(2, 1, 0)$
 $\vec{v}(-1, 0, 3)$



Planaren
 ekuazioa
 bi modutara
 kalkulatu da.

1.) MODUA

$$(x, y, z) = (1, -3, 2) + \lambda(2, 1, 0) + \mu(-1, 0, 3) \quad \text{Bektoren}$$

$$\begin{cases} x = 1 + 2\lambda - \mu \\ y = -3 + \lambda \\ z = 2 + 3\mu \end{cases} \quad \text{Parametrikoki}$$

Inplizitu bidez λ eta μ ezabatu:

$$\begin{cases} 2\lambda - \mu = x - 1 \\ \lambda = y + 3 \\ 3\mu = z - 2 \end{cases} \quad \begin{vmatrix} 2 & -1 & x-1 \\ 1 & 0 & y+3 \\ 0 & 3 & z-2 \end{vmatrix} = 0$$

$$3(x-1) + z-2 - 6(y+3) = 0$$

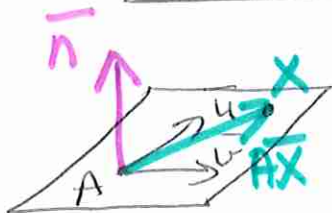
$$3x - 3 + z - 2 - 6y - 18 = 0$$

$$\boxed{\Pi: 3x - 6y + z - 23 = 0}$$

2. MODUA

$$\vec{n} = \vec{u} \wedge \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 0 \\ -1 & 0 & 3 \end{vmatrix} = 3\hat{i} + \hat{k} - 6\hat{j}$$

$$\vec{n} = (3, -6, 1)$$



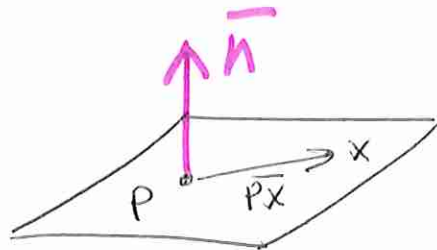
$$\vec{n} \perp \vec{AX} \rightarrow (3, -6, 1) \cdot (x-1, y+3, z-2) = 0$$

$$3(x-1) - 6(y+3) + z-2 = 0$$

$$\vec{AX} = (x-1, y+3, z-2)$$

$$\boxed{\Pi: 3x - 6y + z - 23 = 0}$$

$$b) \pi: \begin{cases} P(2, -3, 1) \\ \bar{n}(5, -3, -4) \end{cases}$$



$$\bar{n} \perp \bar{P}x \rightarrow \bar{n} \cdot \bar{P}x = 0.$$

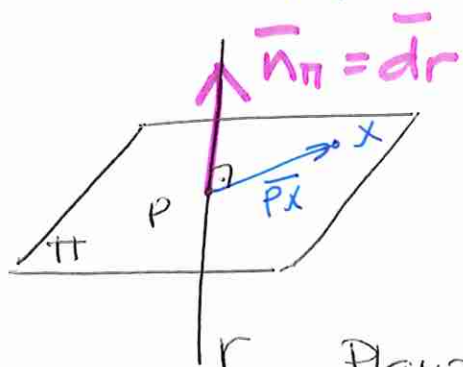
$$\bar{P}x = (x-2, y+3, z-1).$$

$$(5, -3, -4) \cdot (x-2, y+3, z-1) = 0.$$

$$5(x-2) + (-3)(y+3) - 4(z-1) = 0$$

$$\boxed{\pi: 5x - 3y - 4z - 15 = 0}$$

$$c) \begin{cases} \pi \perp r \\ P(1, 0, 1) \end{cases} \quad r: \frac{x}{2} = \frac{y+1}{-1} = \frac{z}{3}.$$



r vektorearen $\bar{d}r$, planoaren
vektore normala da:

$$\bar{d}r = \bar{n}_{\pi}$$

$$\bar{d}r = (2, -1, 3) = \bar{n}_{\pi}.$$

Planoaren ekuazioa kalkulatzeko.

$$\bar{n} \perp \bar{P}x \quad \bar{P}x = (x-1, y, z-1)$$

$$\bar{n} \cdot \bar{P}x = 0$$

$$(2, -1, 3) \cdot (x-1, y, z-1) = 0$$

$$2(x-1) - y + 3(z-1) = 0.$$

$$2x - 2 - y + 3z - 3 = 0.$$

$$\pi: \boxed{2x - y + 3z - 5 = 0}$$

22. EKVADZIO PARAMETRIKAK ETO INPLIZITAK

X4 PLANA

$$\begin{cases} x = 1 \\ y = 1 \\ z = 0 \end{cases}$$

$$z = 0$$

Y2 PLANA

$$\begin{cases} x = 0 \\ y = 1 \\ z = 1 \end{cases}$$

$$x = 0$$

X2 PLANA

$$\begin{cases} x = 1 \\ y = 0 \\ z = 1 \end{cases}$$

$$y = 0$$

23. PLANEN EK. PARAMETRIKAK

a) $z = 3$

$$\begin{cases} x = 1 \\ y = 1 \\ z = 3 \end{cases}$$

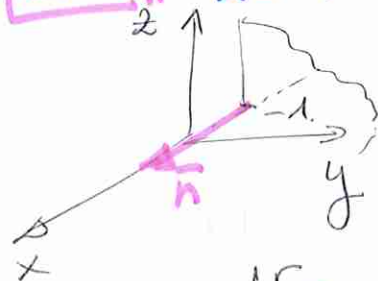
b) $x = -1$

$$\begin{cases} x = -1 \\ y = 1 \\ z = 1 \end{cases}$$

c) $y = 2$

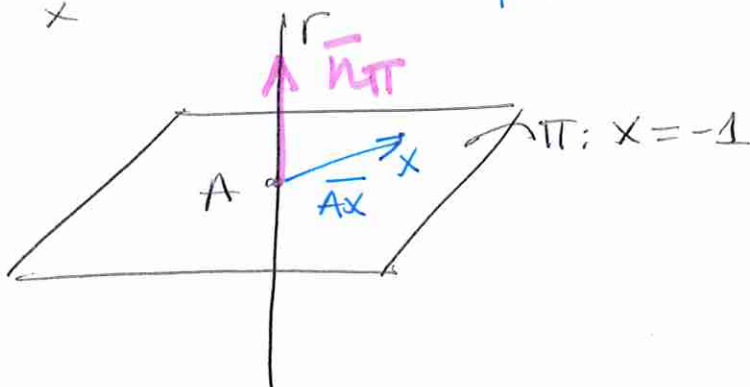
$$\begin{cases} x = 1 \\ y = 2 \\ z = 1 \end{cases}$$

24. $x = -1$ planaren bektore normale.



$$\vec{n} = (1, 0, 0)$$

$r \mid x = -1$ planarekin r perpendikulara
 $A(2, 3, 0)$



Planaren bektore normale, zuzenaren $\vec{dr}$ da.

$$r \mid \begin{cases} A(2, 3, 0) \\ \vec{dr} = (1, 0, 0) \end{cases}$$

$$r: \begin{cases} x = 2 + t \\ y = 3 \\ z = 0 \end{cases} \Rightarrow$$

$$x - 2 = \frac{y - 3}{0} = \frac{z}{0}$$

25 kalkulatu m eta n , paraleloak izateko.

$$\alpha: mx + y - 3z - 1 = 0 \rightarrow \bar{n}_\alpha = (m, 1, -3)$$

$$\beta: 2x + ny - z - 3 = 0 \rightarrow \bar{n}_\beta = (2, n, -1)$$

Izan daitezke koinkidentzak?

PLANOK PARALELOAK IZATEKO $\rightarrow$ Bektore normalak proporzionalak

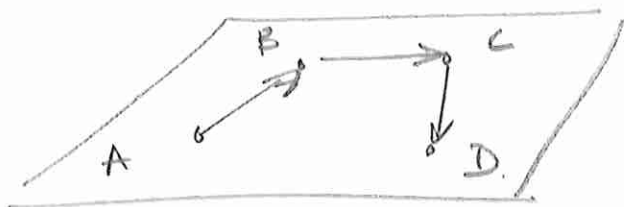
$$\frac{m}{2} = \frac{1}{n} = \frac{-3}{-1} \rightarrow \boxed{\begin{matrix} m=6 \\ n=1/3 \end{matrix}}$$

$$\left\{ \begin{array}{l} \alpha: 6x + y - 3z - 1 = 0 \\ \beta: 2x + \frac{1}{3}y - z - 3 = 0 \end{array} \right\} \leftarrow \text{paraleloak}$$

Paraleloak dira eta ez koinkidentzak
goi gutxiak proporzionalak direlako
goi ordesa izan ekin.

$$\frac{6}{2} = \frac{1}{1/3} = \frac{-3}{-1} \quad \left(\neq \frac{-1}{-3} \right) !!$$

26 $A(0,1,-2)$ $B(1,2,0)$ $C(0,0,1)$ $D(1,0,m)$
kalkulatu m planoakideak izateko



lau puntuen artean 3 bektore
sortu, osatutako matrizearen
heine, 2 izan behar da

$$\overline{AB} = (1, 2, 0) - (0, 1, -2) = (1, 1, 2) \quad \text{Determinante} = 0, \quad 2 \text{ bektore}$$

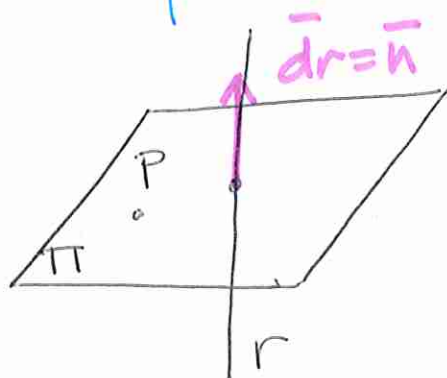
$$\overline{BC} = (0, 0, 1) - (1, 2, 0) = (-1, -2, 1) \quad \text{bektore LI izateko.}$$

$$\overline{CD} = (1, 0, m) - (0, 0, 1) = (1, 0, m-1)$$

$$\begin{vmatrix} 1 & 1 & 2 \\ -1 & -2 & 1 \\ 1 & 0 & m-1 \end{vmatrix} = -2(m-1) + 1 + 4 + m-1 = -2m + 2 + 1 + 4 + m - 1 = -m + 6 = 0$$

$$\boxed{m=6}$$

27. $\left\{ \begin{array}{l} \pi \perp r: \\ \pi \left\{ P(1, -4, 2) \right. \end{array} \right.$



$$r: \begin{cases} x - 2y + 6 = 0 \\ 3x + 2y - 4 = 0. \end{cases}$$

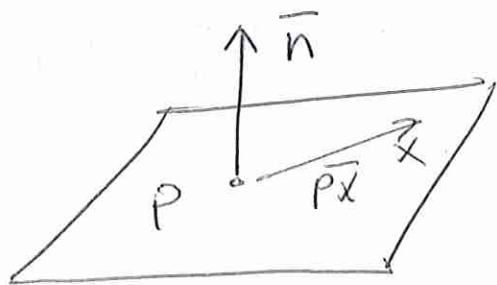
π plano kolkulatuako puntua eta $\bar{n}_\pi$ beko dozu (zuzenoreu uorobide bektorea izango dela)

Zuzenoreu uorobide bektorea ateratu ko implizitetsu

$$r: \begin{cases} x - 2y + 6 = 0 \rightarrow \bar{n}_1 = (1, -2, 0) \\ 3x + 2y - 4 = 0 \rightarrow \bar{n}_2 = (3, 2, 0) \end{cases}$$

$$\bar{dr} = \bar{n}_1 \wedge \bar{n}_2 = (1, -2, 0) \wedge (3, 2, 0) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 0 \\ 3 & 2 & 0 \end{vmatrix} = 8\hat{k} = (0, 0, 8).$$

Planoa kolkulatuako π : $\left\{ \begin{array}{l} P(1, -4, 2) \\ \bar{n}_\pi = \bar{dr} = (0, 0, 8) \end{array} \right.$



$$\bar{n} \perp \overrightarrow{PX} \rightarrow \bar{n} \cdot \overrightarrow{PX} = 0.$$

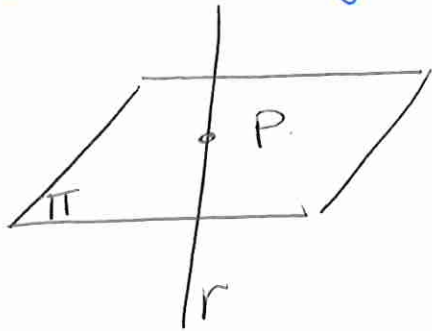
$$\overrightarrow{PX} = (x-1, y+4, z-2)$$

$$(0, 0, 8) \cdot (x-1, y+4, z-2) = 0$$

$$8(z-2) = 0$$

$$\boxed{\pi: z-2=0}$$

28. r: $\frac{x}{2} = y-3 = \frac{z-2}{-3}$ ebakiteu dau $\pi: x-2y+z=1$
 konpustatu eta ebakite puntuak kalkulatu.



r zuzenaren parametroak: $\left. \begin{array}{l} x = 2\lambda \\ y = 3 + \lambda \\ z = 2 - 3\lambda \end{array} \right\} P(2\lambda, 3+\lambda, 2-3\lambda) \in r$
 Ordezkatzen da puntu orokor bat π planuan. $\rightarrow P \in \pi$ probatzeko.

$$2\lambda - 2(3 + \lambda) + 2 - 3\lambda = 1$$

$$2\lambda - 6 - 2\lambda + 2 - 3\lambda = 1$$

$$-3\lambda - 5 = 0$$

$\lambda = -5/3 \rightarrow$ Soluzioa dagoa eta P puntuak ebakitzen dau.

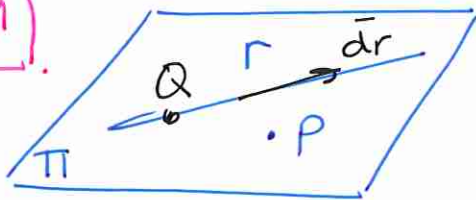
$$x = -10/3$$

$$y = 3 - 5/3 = 4/3$$

$$z = 2 - 3(-5/3) = 2 + 5 = 7$$

$$\rightarrow \boxed{P(-10/3, 4/3, 7)}$$

29.



kalkulatu π parametroak.

$$P(2, 1, 2)$$

$$r: x-2 = \frac{y-3}{-1} = \frac{z-4}{-3}$$

$$\vec{dr} = (1, -1, -3)$$

$$D = (2, 3, 4)$$

Planuaren ekuazioa bertuteko Puntu bat eta bi bektore behar dira

$$Q(2, 3, 4)$$

$$\vec{PQ} = (2, 3, 4) - (2, 1, 2) = (0, 2, 2)$$

$$\vec{dr} = (1, -1, -3)$$

$$\left. \begin{array}{l} x = 2 + \lambda \\ y = 3 - \lambda + 2\mu \\ z = 4 - 3\lambda + 2\mu \end{array} \right\}$$

(30). $r: \frac{x-1}{2} = y = z-2$

$\vec{dr} = (2, 1, 1) \quad P(1, 0, 2)$

$s: \begin{cases} x - 2z = 5 \\ x - 2y = 11 \end{cases}$

$\vec{ds} = \vec{n}_1 \wedge \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -2 \\ 1 & -2 & 0 \end{vmatrix}$
 $\vec{ds} = (-4, -2, -2)$

a) konprobatu Π dirau:

Paraleloak dirau jakiteko atxeratu da ea $\vec{ds}$ eto $\vec{dr}$ proportionalak dirau.

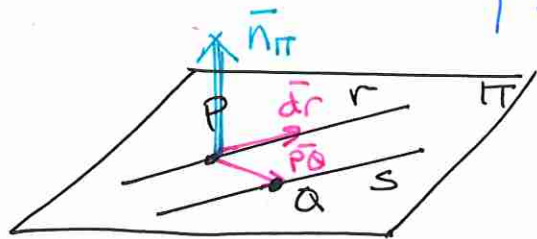
$\vec{dr} = (2, 1, 1)$
 $\vec{ds} = (-4, -2, -2)$

$\frac{2}{-4} = \frac{1}{-2} = \frac{1}{-2} \Rightarrow$ Proportional.
 dirau

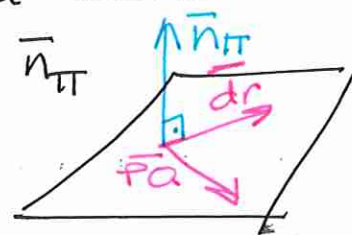
Jakiteko PARALELOAK edo KONTRIDIENTEAK dirau atxeratu da ea P puntua ($P \in r$), S zuzenean dagoan. $P(1, 0, 2) \rightarrow s: \begin{cases} 1 - 2 \cdot 2 \neq 5 \\ 1 - 2 \cdot 0 \neq 11 \end{cases} \rightarrow \underline{\underline{P \notin S}}$

$P \notin S \rightarrow$ ZUZEN PARALELOAK DIRA

b) Π Planoa $\begin{cases} r \in \Pi \\ s \in S \end{cases}$



Planoren bi bektore lotu etken biekiko $\perp$ dau beste bektore bat lotu ariteke $\vec{n}_{\Pi}$ izango da.



Lotzen da bektore bati bat $\vec{PQ}$

$Q \in S \rightarrow x=0 \rightarrow z=-5/2 \rightarrow y=-11/2$

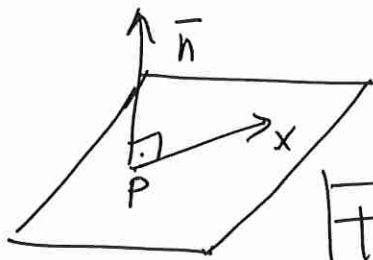
$\vec{PQ} = (0 - \frac{1}{2}, -\frac{5}{2}) - (1, 0, 2) = (-\frac{1}{2}, -\frac{11}{2}, -\frac{9}{2})$

$\vec{n}_{\Pi} = \vec{PQ} \wedge \vec{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1/2 & -11/2 & -9/2 \\ 2 & 1 & 1 \end{vmatrix} = -\frac{1}{2}\hat{i} - 9\hat{j} - \hat{k} + 11\hat{k} + \hat{j} + \frac{9}{2}\hat{i}$

$\vec{n}_{\Pi} = \underline{\underline{(-1, -8, 10)}}$

$= -\hat{i} - 8\hat{j} + 10\hat{k}$

$$\vec{n} \perp \vec{PX} \rightarrow \vec{n} \cdot \vec{PX} = 0$$



$$(-1, -8, 10) \cdot (x-1, y, z-2) = 0$$

$$-x+1-8y+10z-20=0$$

$$\Pi: -x-8y+10z-19=0$$

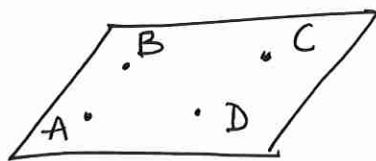
31) Planoan dagot A, B, C, eto D?

$$A(1, 0, 0)$$

$$B(0, 1, 0)$$

$$C(2, 1, 0)$$

$$D(-1, 2, 1)$$



Atiraten dira
u puntuak 3 bektore
eta atiraten da
euren arteko matritesra
heine.

$$\vec{AD} = (-1, 2, 1) - (1, 0, 0) = (-2, 2, 1)$$

$$\vec{AB} = (0, 1, 0) - (1, 0, 0) = (-1, 1, 0)$$

$$\vec{CD} = (-1, 2, 1) - (2, 1, 0) = (-3, 1, 1)$$

$$\begin{vmatrix} -2 & 2 & 1 \\ -1 & 1 & 0 \\ -3 & 1 & 1 \end{vmatrix} = -2(-1+3) = 2 \neq 0$$

$$\text{rank}(M) = 3 \rightarrow 3 \text{ bektore LI}$$

et dira
planokideak

32

TIKRU PLANDEN ARTIKO POSIZIA

10

$$a) \begin{cases} x + 2y - z - 3 = 0 \\ 3y + 2z - 1 = 0 \\ x + y + z - 2 = 0 \end{cases}$$

$$M' = \begin{pmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 3 & 2 & | & 1 \\ 1 & 1 & 1 & | & 2 \end{pmatrix}$$

n

$|M|=8$ $\text{heiu}(n) = \text{heiu}(n') = 3$

TIKRU PLANDEN PUNTA BATAN EBAKITEN DIRA.

$$b) \begin{cases} 2x - y + z = 3 \\ x - y + z = 2 \\ 3x - y + z = 4 \end{cases}$$

$$M' = \begin{pmatrix} 2 & -1 & 1 & | & 3 \\ 1 & -1 & 1 & | & 2 \\ 3 & -1 & 1 & | & 4 \end{pmatrix}$$

$n \rightarrow \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} = -2 + 1 = -1$

$\begin{vmatrix} 2 & -1 & 1 \\ 1 & -1 & 1 \\ 3 & -1 & 1 \end{vmatrix} = -2 - 3 - 1 + 3 + 1 + 2 = 0.$

$n' \rightarrow \begin{vmatrix} 2 & -1 & 3 \\ 1 & -1 & 2 \\ 3 & -1 & 4 \end{vmatrix} = -8 - 3 + 9 + 4 + 4 = 0.$

$\left\{ \begin{array}{l} \text{heiu}(n) = \text{heiu}(n') \\ = 2 \\ \text{Bat indeterm.} \\ \text{PLANDAK} \\ \text{EUKAN} \\ \text{EBAKITEN} \\ \text{DIRA} \\ \text{ZULEN BATAN} \end{array} \right.$

$$c) \begin{cases} x - y + z - 1 = 0 \\ 3x + y - 2z = 0 \\ 2x + 2y - 3z + 4 = 0 \end{cases}$$

$$M' = \begin{pmatrix} 1 & -1 & 1 & | & 1 \\ 3 & 1 & -2 & | & 0 \\ 2 & 2 & -3 & | & -4 \end{pmatrix}$$

n

$n \rightarrow \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} = 1 + 3 = 4$

$\begin{vmatrix} 1 & -1 & 1 \\ 3 & 1 & -2 \\ 2 & 2 & -3 \end{vmatrix} = -3 + 4 + 6 - 2 - 9 + 4 = 0$

$n' \rightarrow \begin{vmatrix} 1 & -1 & 1 \\ 3 & 1 & 0 \\ 2 & 2 & -4 \end{vmatrix} = -12 \neq 0$

$\left\{ \begin{array}{l} \text{heiu}(n) = 2 \neq \text{heiu}(n') = 3 \\ \text{PLANDAK BINAN} \\ \text{BINAN EBAKITEN DIRA} \end{array} \right.$

Batorcelio



$$d) \begin{cases} x+y-2z = 0 \\ -x+y+z = -1 \\ 2x-z = -2 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ -1 & 1 & 1 & -1 \\ 2 & 0 & -1 & -2 \end{array} \right) \quad \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2$$

$$\begin{vmatrix} 1 & 1 & -2 \\ -1 & 1 & 1 \\ 2 & 0 & -1 \end{vmatrix} = -1 + 2 + 4 - 1 \neq 0$$

$$\begin{vmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 2 & 0 & -2 \end{vmatrix} = -2 - 2 - 2 + 1 = -5$$

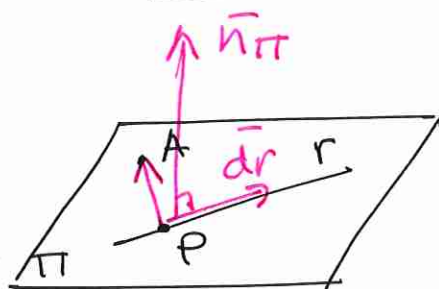
kein $\pi = 3$. = kein π'

Planoak puntu batzuek ebakitzu dira

33

$$A(1,0,1)$$

$$r: \begin{cases} x+y-z+1=0 \\ 2x-y+2z=0 \end{cases}$$



- Planooren ek. $\bar{u}_{\pi}$ eto Puntu bat behar da.
- $\bar{u}_{\pi}$ kalkulatzeko planoaren bi bektore behar dira. $\bar{dr}$ eta $\bar{PA}$, non P r-ren puntu bat dau.
- $\bar{dr}$ ateratzeko r-ren inplizitatuak:

$$\begin{aligned} \bar{dr} &= \bar{u}_1 \wedge \bar{u}_2 = (1, 1, -1) \wedge (2, -1, 2) = \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -1 & 2 \end{vmatrix} = 2\hat{i} - 2\hat{j} - \hat{k} - 2\hat{k} - 2\hat{j} - \hat{i} = \\ &= \hat{i} - 4\hat{j} - 3\hat{k} = \underline{\underline{(1, -4, -3)}} = \bar{dr} \end{aligned}$$

- P puntua:

$$\begin{array}{rcl} x=0 & y-z+1=0 & \\ & -y+2z=0 & \\ \hline & z+1=0 & z=-1 \quad y=-2 \end{array}$$

$$P(0, -2, -1)$$

$$\bar{PA} = (1, 0, 1) - (0, -2, -1) = \underline{\underline{(1, 2, 2)}} = \bar{PA}$$

• $\bar{n}_\pi$ kalkulatuks:

$$\bar{n}_\pi = \overrightarrow{PA} \wedge \overrightarrow{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 1 & -4 & -3 \end{vmatrix} =$$

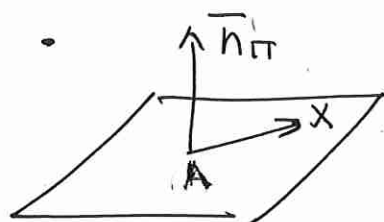
$$= -6\hat{i} + 2\hat{j} - 4\hat{k} - 2\hat{k} + 3\hat{j} + 8\hat{i} =$$

$$= 2\hat{i} + 5\hat{j} - 6\hat{k}$$

$$\bar{n}_\pi = (2, 5, -6)$$

Produkta mistos

$$\begin{array}{l} \overrightarrow{AX} \\ \overrightarrow{PA} \\ \overrightarrow{dr} \end{array} \begin{vmatrix} x-1 & y & z-1 \\ 1 & 2 & 2 \\ 1 & -4 & -3 \end{vmatrix}$$



$$A(1, 0, 1)$$

$$\bar{n} \perp \overrightarrow{AX} \rightarrow \bar{n} \cdot \overrightarrow{AX} = 0.$$

$$(2, 5, -6) \cdot (x-1, y, z-1) = 0$$

$$2(x-1) + 5y - 6(z-1) = 0.$$

$$2x - 2 + 5y - 6z + 6 = 0$$

$$\boxed{\pi: 2x + 5y - 6z + 4 = 0.}$$

34. Γ eta π -ren posizio ERLATIBOA:

$$\Gamma: \begin{cases} x = -t \\ y = 0 \\ z = 1+t \end{cases} \quad t \in \mathbb{R}$$

$$\pi: 2x - y + 3z = 6.$$

$$\bar{n}_\pi = (2, -1, 3)$$

$$\overrightarrow{dr} = (-1, 0, 1)$$

$$\overrightarrow{dr} \cdot \bar{n}_\pi = (-1, 0, 1) \cdot (2, -1, 3) = -2 + 0 + 3 = 1 \neq 0.$$

$\overrightarrow{dr} \cdot \bar{n}_\pi \neq 0$ dauzat Γ eta π ez dira paraleloak.
beraz EBAKITZEN dira.

35

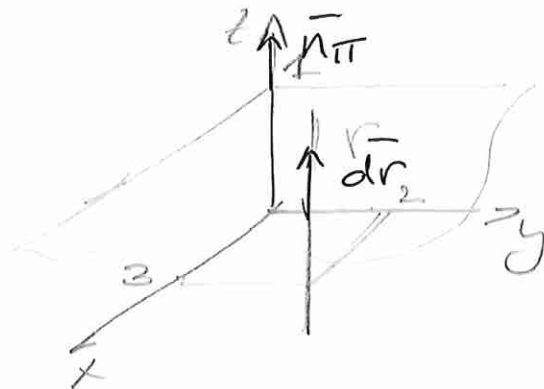
Posizio ERLATIBOA.

$$r: \begin{cases} x=3 \\ y=2 \end{cases}$$

$$\pi: z=1$$

$$\bar{n}_\pi = (0, 0, 1)$$

$$\bar{dr} = (0, 0, 1)$$



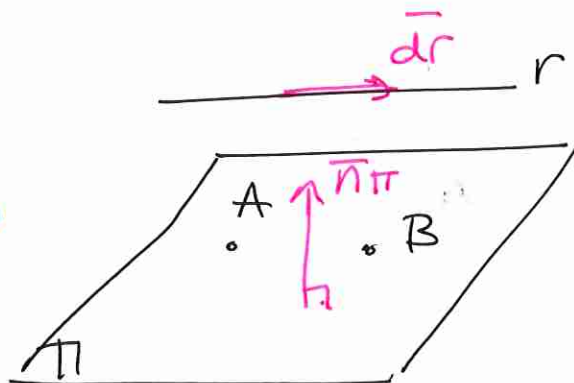
Zuzena eta planoa perpindikularrok dira eta ebakitzeko puntua $(3, 2, 1)$ da.

36

$$r: \begin{cases} 3x - 2y + 1 = 0 \\ 2y + 3z - 3 = 0 \end{cases}$$

$$A(1, -3, 2)$$

$$B(0, 1, 1)$$



Planoa aterotzeko puntua eta $\bar{n}_\pi$ behar da

$$\bar{dr} = (3, -2, 0) \wedge (0, 2, 3) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 0 \\ 0 & 2 & 3 \end{vmatrix} =$$

$$= -6\hat{i} + 6\hat{k} - 9\hat{j} = (-6, -9, 6) // (2, 3, -2)$$

$$\bar{u}_\pi = \bar{dr} \wedge \bar{AB}$$

$$\bar{AB} = (0, 1, 1) - (1, -3, 2) = (-1, 4, -1)$$

Bideko matrikoa

$$\begin{vmatrix} \bar{AX} & \bar{dr} & \bar{AB} \\ x-1 & y+3 & z-2 \\ 2 & 3 & -2 \\ -1 & 4 & -1 \end{vmatrix}$$

$$\begin{aligned} \rightarrow \bar{u}_\pi = \bar{dr} \wedge \bar{AB} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -2 \\ -1 & 4 & -1 \end{vmatrix} = -3\hat{i} + 2\hat{j} + 8\hat{k} \\ &+ 3\hat{k} + 2\hat{j} + 8\hat{i} \\ &= 5\hat{i} + 4\hat{j} + 11\hat{k} \\ &= \underline{\underline{(5, 4, 11)}} \end{aligned}$$

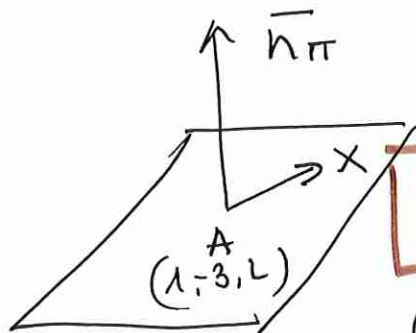
Planoa aterotzeko:

$$\bar{n} \cdot \bar{AX} = 0$$

$$(5, 4, 11) \cdot (x-1, y+3, z-2) = 0$$

$$5x - 5 + 4y + 12 + 11z - 22 = 0$$

$$\pi: \underline{\underline{5x + 4y + 11z - 15 = 0}}$$



37

kalkulatu m plauzak paraleloak eta
perpendikularak izateko.

$$\pi: mx + 2y - 3z - 1 = 0$$

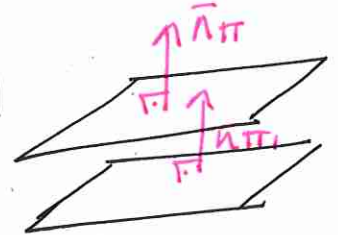
$$\pi': 2x - 4y + 6z + 5 = 0$$

$$\bar{n}_{\pi} = (m, 2, -3)$$

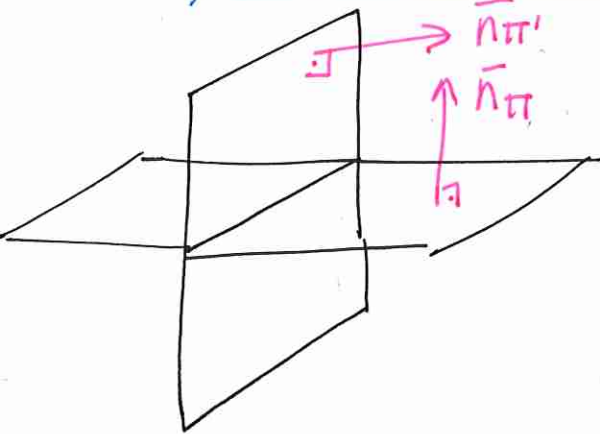
$$\bar{n}_{\pi'} = (2, -4, 6)$$

a) PARALELOAK

Paraleloak izateko $\bar{n}_{\pi} = \alpha \bar{n}_{\pi'}$ proporzionalak izan
behar dira. $\frac{m}{2} = \frac{2}{-4} = \frac{-3}{6} \rightarrow \boxed{m = -1}$



b) PERPENDIKULARAK



Plauzak perpendikularak
badira $\bar{n}_{\pi} \perp \bar{n}_{\pi'}$, beraz

$$\bar{n}_{\pi} \cdot \bar{n}_{\pi'} = 0.$$

$$(m, 2, -3) \cdot (2, -4, 6) = 0.$$

$$2m - 8 - 18 = 0$$

$$2m - 26 = 0$$

$$\boxed{m = 13}$$

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$$r: \begin{cases} x = 1+t \\ y = -1+2t \\ z = t \end{cases}$$

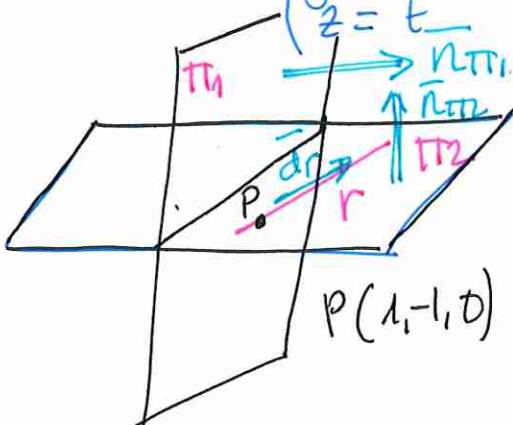
$$\bar{dr} = (1, 2, 1)$$

$$\pi_1: 2x + y - z - 2 = 0 \quad \bar{n}_1 = (2, 1, -1)$$

kalkulatu π_2

$$\pi_2 \perp \pi_1.$$

$$\bar{n}_2 \perp \bar{n}_1$$



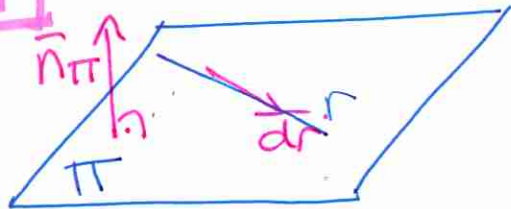
$$P(1, -1, 0) \quad \bar{n}_2 = \bar{dr} \wedge \bar{n}_1 \quad \left. \begin{aligned} \bar{r} \cdot (\bar{dr} \wedge \bar{n}_1) &= 0. \\ \bar{r} \cdot \bar{n}_2 &= 0. \end{aligned} \right\}$$

$$\begin{vmatrix} \bar{r} \cdot \bar{dr} & \bar{r} \cdot \bar{n}_1 \\ x-1 & y+1 & z \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0 \Rightarrow$$

$$-2(x-1) + 2(y+1) + z - 4z + y + 1 - x - 1 = 0$$

$$\boxed{\pi_2: -3x + 3y - 3z + 6 = 0}$$

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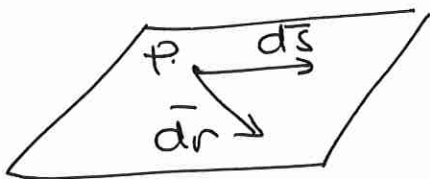


$$S: \begin{cases} x-y-z=0 \\ x-2y+z=0 \end{cases}$$

$$r: \frac{x-1}{2} = y = 2-z \quad P(1,0,2)$$

$$\begin{aligned} \vec{dS} &= (1, -1, -1) \wedge (1, -2, 1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & -2 & 1 \end{vmatrix} = \\ &= -\hat{i} - \hat{j} - 2\hat{k} + \hat{k} - \hat{j} - 2\hat{i} = -3\hat{i} - 2\hat{j} - \hat{k} \\ &= (-3, -2, -1) \parallel \underline{\underline{(3, 2, 1)}} \end{aligned}$$

$$\vec{dr} = \underline{\underline{(2, 1, -1)}}$$



Planaaren ek aterotrekko
bi bektore eto $P(1,0,2)$

$$\begin{vmatrix} \vec{Px} & \vec{dS} & \vec{dr} \\ x-1 & y & z-2 \\ 3 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0.$$

$$\begin{aligned} -2(x-1) + 2y + 3(z-2) - 4(z-2) + 3y - (x-1) &= 0 \\ -3x + 3 + 5y - z + 2 &= 0 \end{aligned}$$

$$\Pi: -3x + 5y - z + 5 = 0$$

• Beste modura. $\vec{n}$ eto $P(1,0,2)$.

$$\vec{u} = \vec{dS} \wedge \vec{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = -3\hat{i} + 5\hat{j} - \hat{k}$$

$$\vec{u} \perp \vec{Px} = 0.$$

$$(-3, 5, -1) \cdot (x-1, y, z-2) = 0 \Rightarrow$$

